

(we might do it in a dedicated FastTrack someday!), but here's what you need to know. While everyday computers work on 0s and 1s, a quantum computer works on 0s, 1s, and a mixture of 0 and 1. And that is why it is difficult for people to wrap their heads around the idea – not to mention scientists, who've yet to create anything that can call itself a quantum computer without a smirk. But, if a manifestation does see the light of day, here's what it promises to establish – the vast majority of the problems that computer science thinks are hard, because of the exponential nature of problems, can theoretically be converted into classical, polynomial sets. So instead of 2^{256} , we get 2^{256} . And we can confidently vouch for the fact that it is much, much less than the number of atoms in the Universe. Which is why it would take anything between 10-20 seconds for one instance of such a computer to bring the entire cryptography system to its knees, safety and security online would be a lie.

Hold your horses there, soldier! You don't need to march to the bank to withdraw your hard-earned cash just yet. There are no quantum computers in existence, as of yet. At least none that we know of (though you never know with CIA). In fact, the furthest that people have been about it, is barely past the start line. So you probably have nothing to worry about. For now, that is. If you ever hear of a quantum computer being built, you know what you need to do.

Now that the groundwork for generic model of online transactions has been established, we should get you started on acquainting yourself with some of the real world implementations for this.

Starting off, we think you've probably heard of...

PayPal

PayPal is, as it has been for some time now, the world's favourite "middleman" in online payment market (yeah, it's a market – a huge market, so to speak). It bases its working model on trust. And to be honest, having been around for so long, all the trust that it gets is fairly commanded. PayPal works by pairing an e-mail id with an account information. So these are

